A Short Tour of Type Classes

A type class names a set of operations. A type becomes an instance of the class by providing them.

Eq gives equality, Ord gives ordering, and Show turns values into text.

Instances can be derived for simple data types, so most code never writes them by hand.

Classes with laws

Some classes come with laws that instances should obey. Monoid instances must be associative and respect the identity.

Functor instances must preserve identity and composition: fmap id equals id.

The compiler does not check these laws, so tests usually do.